

Volume 14

Cyber Security Intelligence Layer (CSIL)

Version: 1.0

Status: Draft

Chapter 1

Vision

The Cyber Security Intelligence Layer transforms raw observations into actionable intelligence.

Instead of asking:

"What vulnerabilities exist?"

It asks:

- What is happening?
 - Why is it happening?
 - How are assets connected?
 - What should happen next?
 - What is the business impact?
 - How confident are we?
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Chapter 2

Intelligence Pipeline

Every observation follows the same lifecycle.

Observation

This ensures consistency across all modules.

Chapter 3

Unified Object Model

Everything in Project Sentinel is an object.

Examples include:

- Website
- Domain
- API
- Server
- Laptop
- Mobile Device
- User
- Cloud Account
- Kubernetes Cluster
- Certificate
- Vulnerability
- Incident
- IOC
- Malware Sample
- Threat Campaign
- Report
- Evidence
- Mission
- Case

Each object has:

- Unique Object ID
- Owner
- Relationships
- Timeline
- Trust Profile
- Risk Profile
- Evidence
- Audit History
- Tags
- Security DNA
- Lifecycle State

Chapter 4

Object Relationships

Instead of isolated findings:

Employee

A vulnerability in one object automatically reveals potential downstream impact.

Chapter 5

Intelligence Correlation

Project Sentinel correlates observations from different engines.

Example:

Discovery Engine

+

Web Security Engine

+

Endpoint Security Engine

+

Cloud Security Engine

+

Threat Intelligence

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Combined Assessment

This provides richer context than isolated scan results.

Chapter 6

Knowledge Graph

The Knowledge Graph is the platform's central reasoning model.

It stores relationships—not just data.

Example relationships:

- Uses

- Owns
- Connects To
- Depends On
- Hosts
- Exposes
- Protects
- Authenticates
- Trusts
- Generates
- Contains
- Remediates

The graph enables efficient impact analysis and navigation.

Chapter 7

Decision Intelligence

The platform prioritizes actions based on:

- Risk
- Business Criticality
- Ease of Remediation
- Dependencies
- Threat Context
- Confidence
- Operational Impact

This supports evidence-based decision making.

Chapter 8

Mission Planner

Instead of isolated scans:

The user defines a mission.

Examples:

- Secure My Website
- Audit My Company
- Investigate Suspicious Activity
- Prepare for ISO Audit
- Reduce External Attack Surface
- Review Mobile Devices
- Perform Authorized Bug Bounty Assessment

The platform assembles the required engines automatically.

Chapter 9

Verification Loop

Every recommendation ends with verification.

Finding

A task is not complete until verification confirms the intended outcome.

Chapter 10

Continuous Learning

Project Sentinel improves its understanding of the environment over time by learning from:

- Approved configurations.
- Historical changes.
- Asset relationships.
- Previous investigations.
- Confirmed incidents.
- Successful remediations.

Learning is transparent and based on evidence—not opaque AI behavior.

PS-ADR-0011

Intelligence as a Platform Capability

Status: Accepted

Decision

Intelligence shall be a platform-wide capability rather than an isolated feature of individual modules.

Rationale

Centralized correlation improves consistency, reduces duplication, and enables explainable reasoning across the platform.

New Core Engine - Mission Orchestrator

One capability I believe will become the "brain" of Project Sentinel is the **Mission Orchestrator**.

Responsibilities include:

- Selecting the appropriate engines.
- Scheduling tasks.
- Managing dependencies.
- Tracking mission progress.
- Handling failures.
- Coordinating verification.
- Producing the final mission report.

Every scan, audit, investigation, or compliance review becomes a managed mission.

New Core Engine - Evidence Correlation Engine (ECE)

The Evidence Correlation Engine combines evidence from all platform modules.

Rather than evaluating each observation independently, it asks:

- Do multiple engines support this conclusion?
- Are there conflicting observations?
- Has this behavior occurred before?
- Which evidence is strongest?

Each conclusion is assigned a confidence level based on the quality and consistency of the evidence.

New Core Engine - Policy Engine

Every organization defines policies.

Examples:

- All administrator accounts must use MFA.
- Production systems must support TLS 1.3.
- Public APIs must require authentication.
- Endpoints must have disk encryption enabled.
- Mobile devices must be protected with a screen lock.

The Policy Engine continuously evaluates the environment against these policies and reports compliance, deviations, and recommended actions.

Project Sentinel Design Principle

I would like to introduce what I think could become the platform's most important engineering principle:

"One Platform - Many Missions"

Instead of developing separate products for:

- Website Security
- Mobile Security
- Endpoint Security

- Cloud Security
- SOC
- Compliance
- Bug Bounty
- Asset Management
- Incident Response

we build **one platform** where these are different missions powered by the same object model, evidence engine, knowledge graph, and workflow system.

Long-Term Vision: Sentinel OS

I want to share one final architectural thought—not for version 1, but as a direction.

If we continue with this architecture, Project Sentinel could eventually evolve into what I would call **Sentinel OS**.

Not an operating system in the traditional sense, but an **operating system for cybersecurity operations**.

Its responsibilities would include:

- Discovering assets.
- Understanding relationships.
- Assessing trust.
- Detecting change.
- Prioritizing work.
- Guiding remediation.
- Verifying outcomes.
- Preserving evidence.
- Supporting compliance.
- Coordinating investigations.
- Assisting analysts.

Everything revolves around a single source of truth.

One additional principle

As we move from design to implementation, I want us to adopt a strict rule:

If we cannot explain how a feature works, how it will be implemented in Python, how it will be tested, and how it will be maintained over the next ten years, it does not become part of the core platform.

That discipline will help Project Sentinel remain ambitious, technically grounded, and maintainable—qualities that are essential if the goal is to build a platform that can one day be trusted by individuals, enterprises, financial institutions, and government organizations.

====[USER]=====

Good and continue

====[ASSISTANT]=====

Excellent. I think we are now entering what will become the **heart of Project Sentinel**.

Everything we have designed until now has prepared us for this.

From this point forward, we stop thinking about scanners.

We stop thinking about pentest tools.

We stop thinking about SOC.

Instead, we think about **Cyber Operations**.

That is a much bigger vision.

PROJECT SENTINEL BLUEPRINT

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